Indian Institute of Technology, Bhilai

Department of Computer Science and Engineering

CSL351: Computer Networks

Assignment 9

Network Layer - Routing

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Part A: Topology Design

A.1 Network Overview

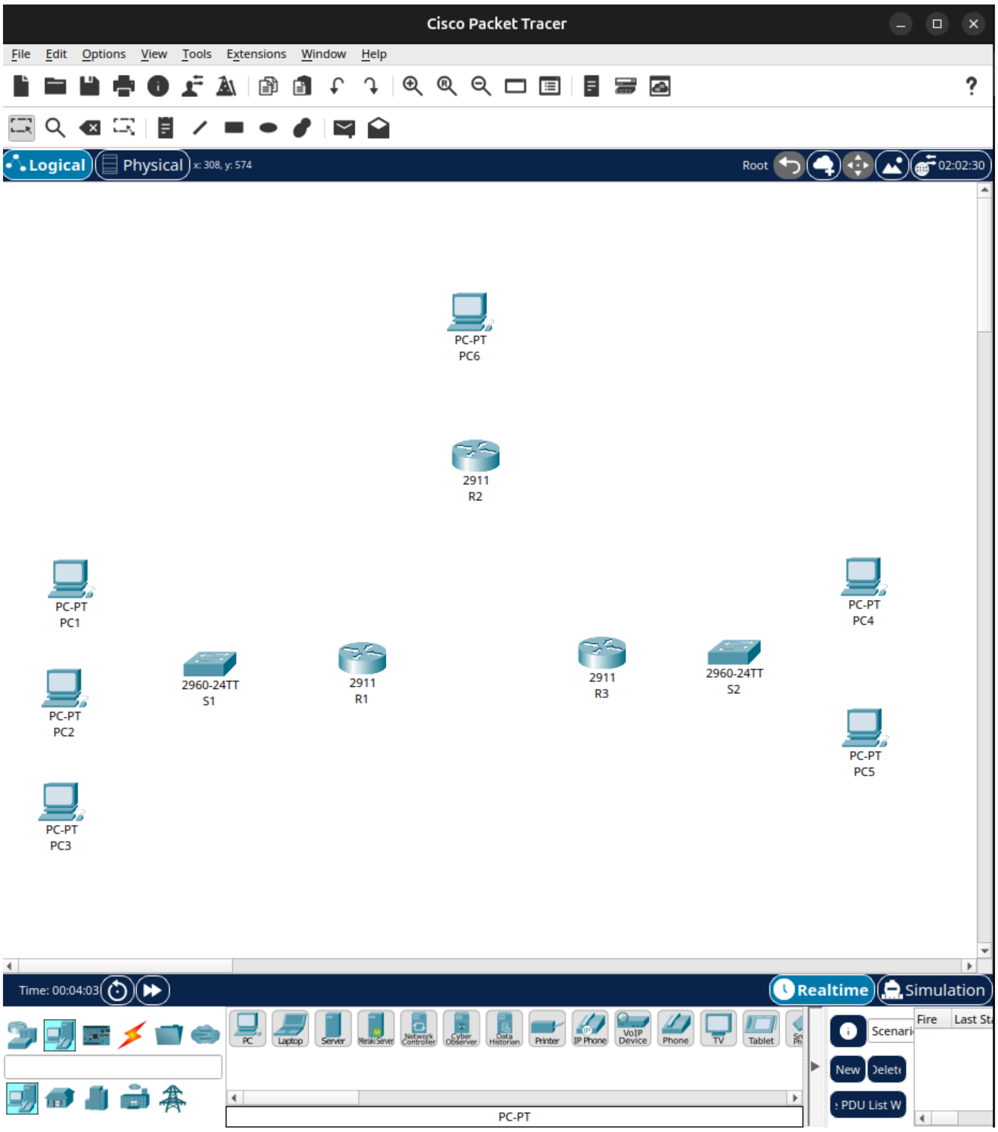
The network consists of 3 routers (R1, R2, R3) connected in a full-mesh (triangle) topology, 2 switches (S1, S2), and 6 end devices (PC1–PC6). The overall structure is:

- R1 ↔ R2 (WAN link)
- R2 ↔ R3 (WAN link)
- R3 ↔ R1 (WAN link)
- R1 → S1 → PC1, PC2, PC3 (LAN 1)
- R3 → S2 → PC4, PC5 (LAN 2)
- R2 → PC6 (direct connection)

A.2 Device Placement

The following devices were placed on the Cisco Packet Tracer workspace:

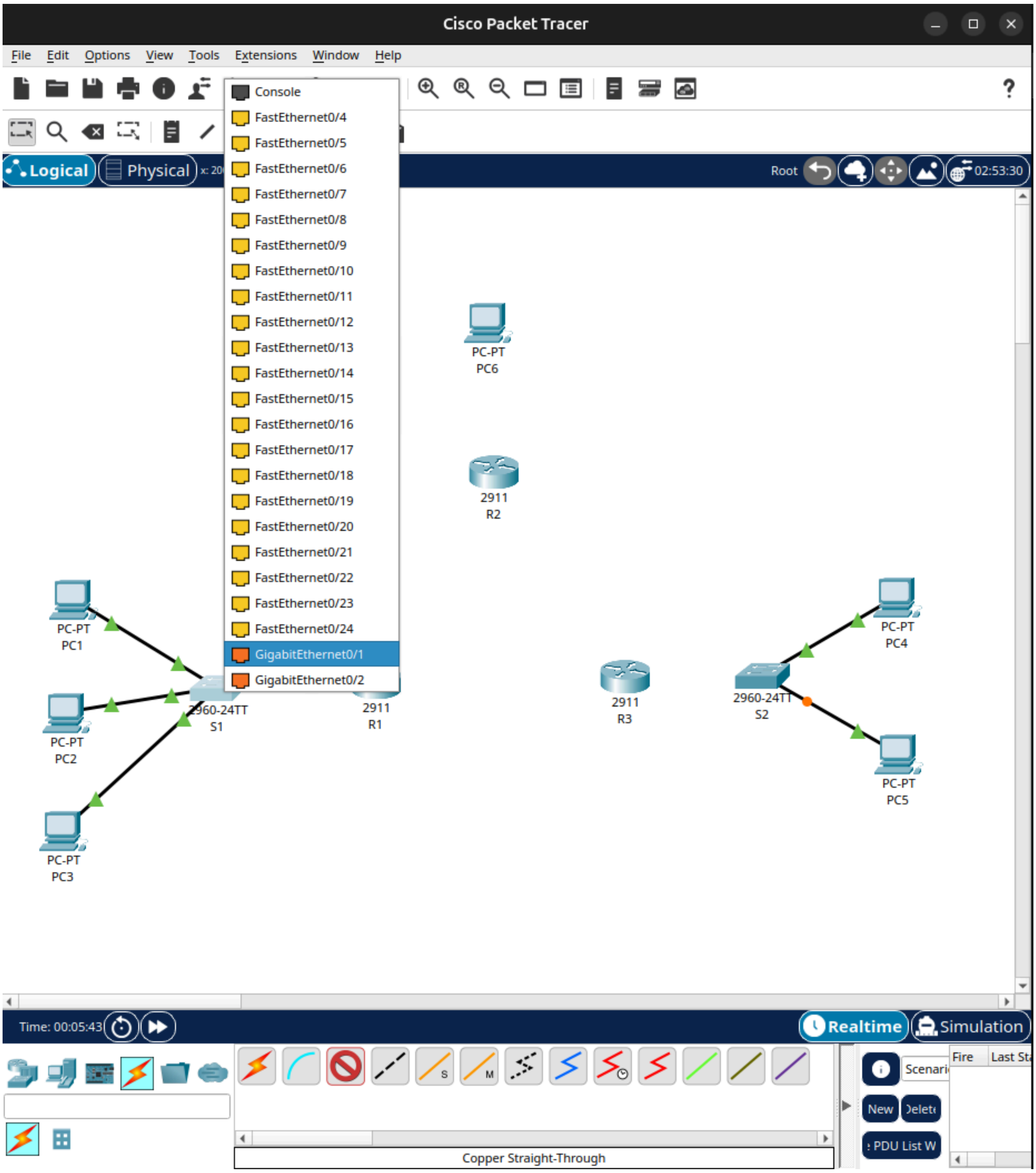
- 3 × Cisco 2911 Routers – R1, R2, R3
- 2 × Cisco 2960 Switches – S1 (connected to R1), S2 (connected to R3)
- 6 × Generic PCs – PC1, PC2, PC3, PC4, PC5, PC6

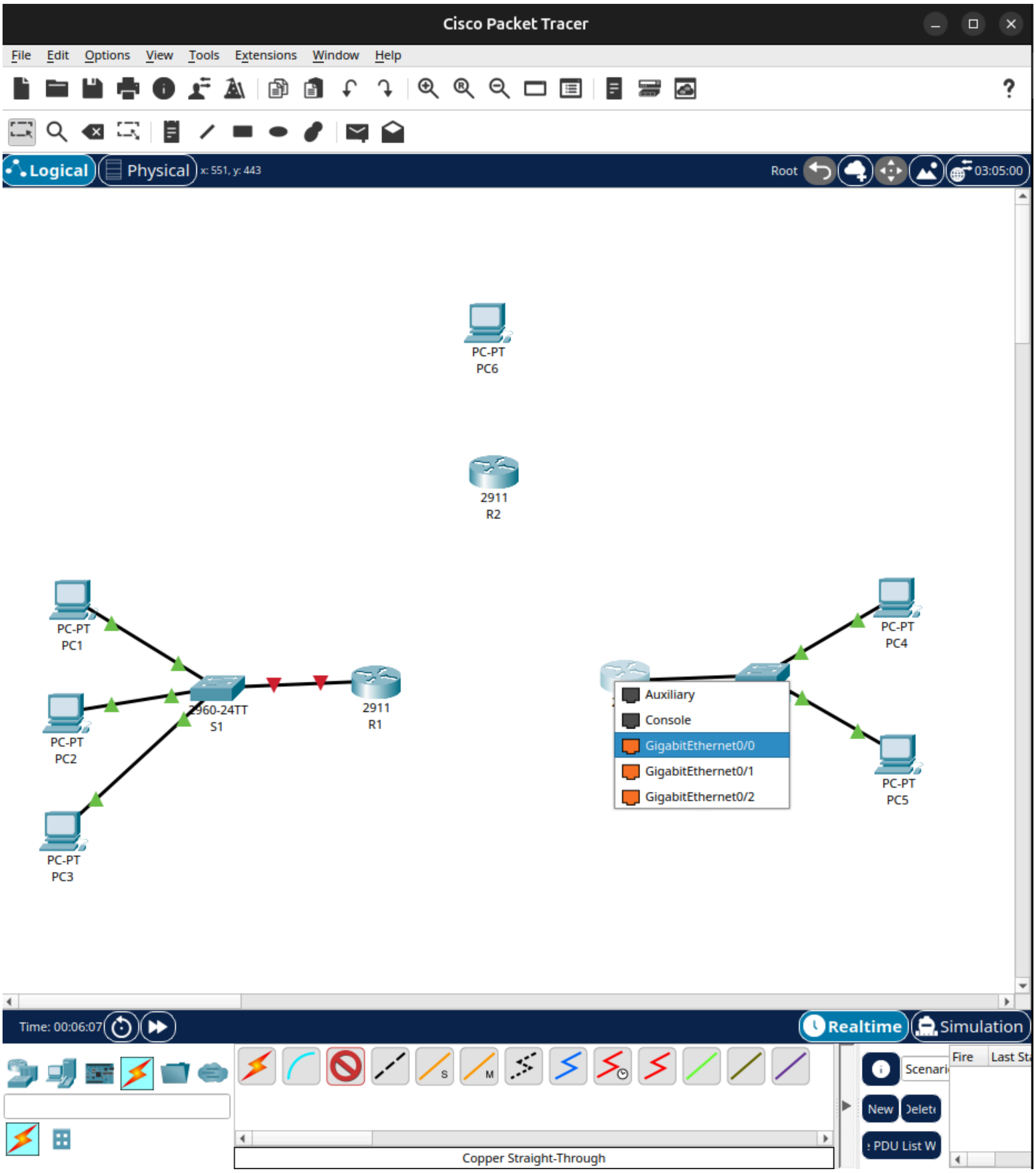


A.3 Cable Connections

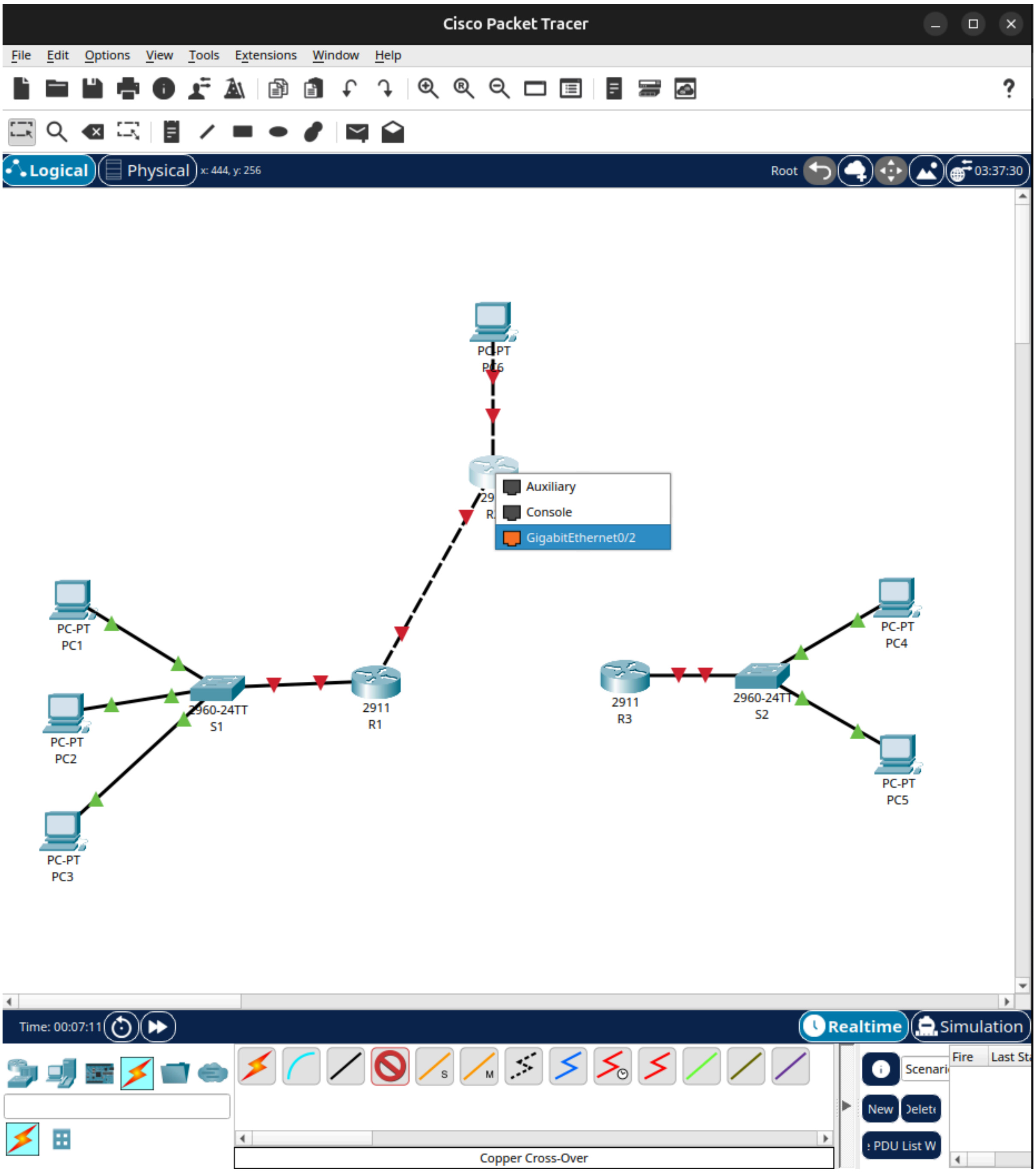
The connections were established according to the assigned network structure:

- **WAN Connections:** The three routers (R1, R2, R3) were connected in a full mesh triangle using Gigabit links. Since these are router-to-router connections, Copper Cross-Over cables were used.
- **LAN Connections:** PCs 1, 2, and 3 were connected to Switch S1, and PCs 4 and 5 were connected to Switch S2 using Copper Straight-through cables . S1 was connected to R1, and S2 was connected to R3 .
- **Direct Connection:** PC6 was connected directly to Router R2 via a Gigabit link.

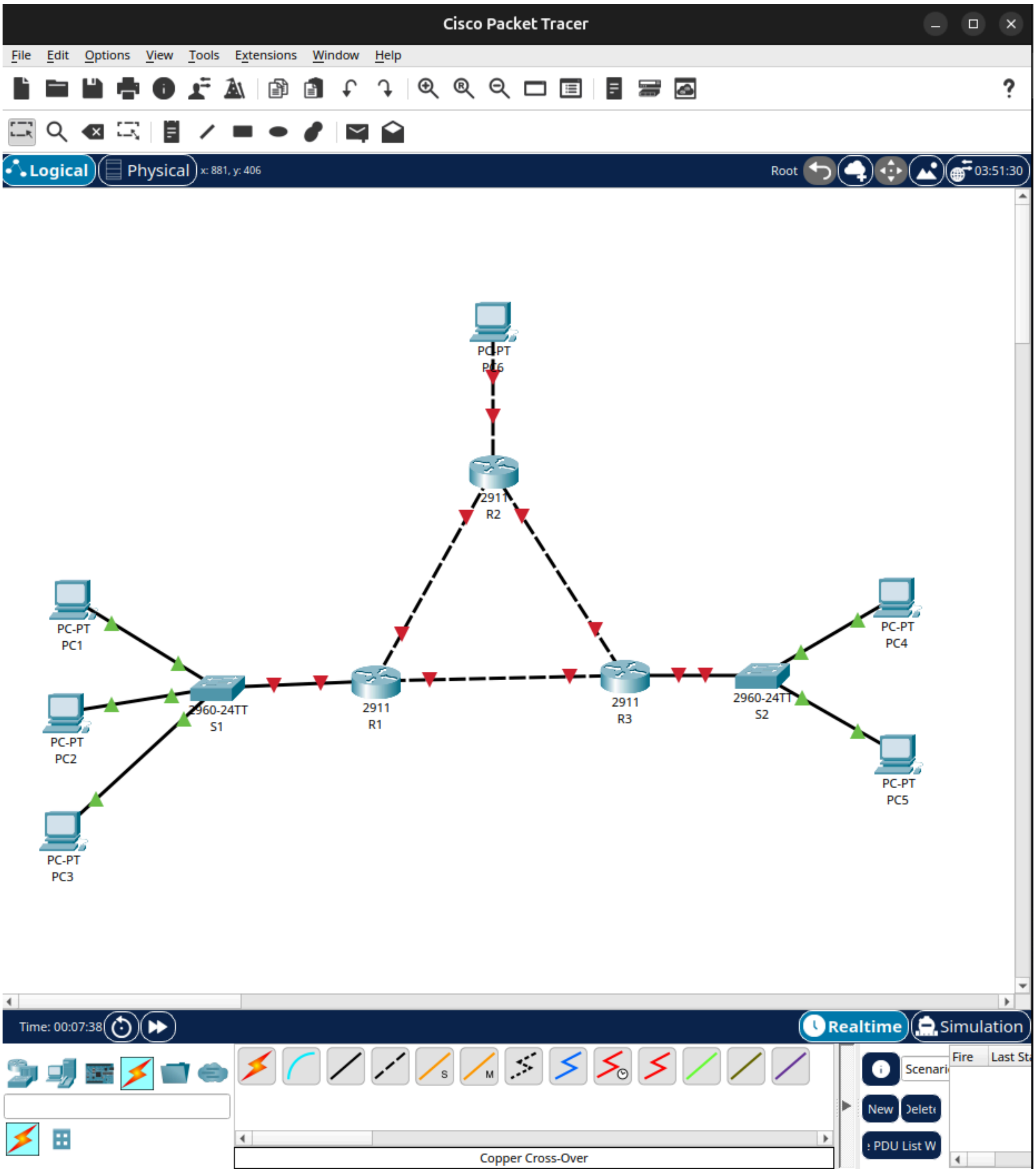




Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 551, y: 443
Root
03:05:00
PC-PT
PC6
2911
R2
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
2960-24TT
S1
2911
R1
PC-PT
PC4
PC-PT
PC5
Auxiliary
Console
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Time: 00:06:07
Realtime
Simulation
Copper Straight-Through
New
PDU List W
Fire


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 444, y: 256
Root
03:37:30
PC-PT
PC6
Auxiliary
Console
GigabitEthernet0/2
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
2960-24TT
S1
2911
R1
2911
R3
2960-24TT
S2
PC-PT
PC4
PC-PT
PC5
Time: 00:07:11
Realtime
Simulation
Copper Cross-Over
Fire
New
PDU List W

A.4 Network Topology Diagram



Part B: Configuration

B.1 IP Address Allocation

The following IP address scheme is used. Four distinct subnets are defined:

- **LAN 1 (R1 & S1):** 192.168.1.0 / 24
- **LAN 2 (R3 & S2):** 192.168.2.0 / 24
- **LAN 3 (R2 direct):** 192.168.3.0 / 24
- **WAN Links:** 10.0.12.0 / 24 (R1-R2), 10.0.23.0 / 24 (R2-R3), and 10.0.31.0 / 24 (R3-R1)

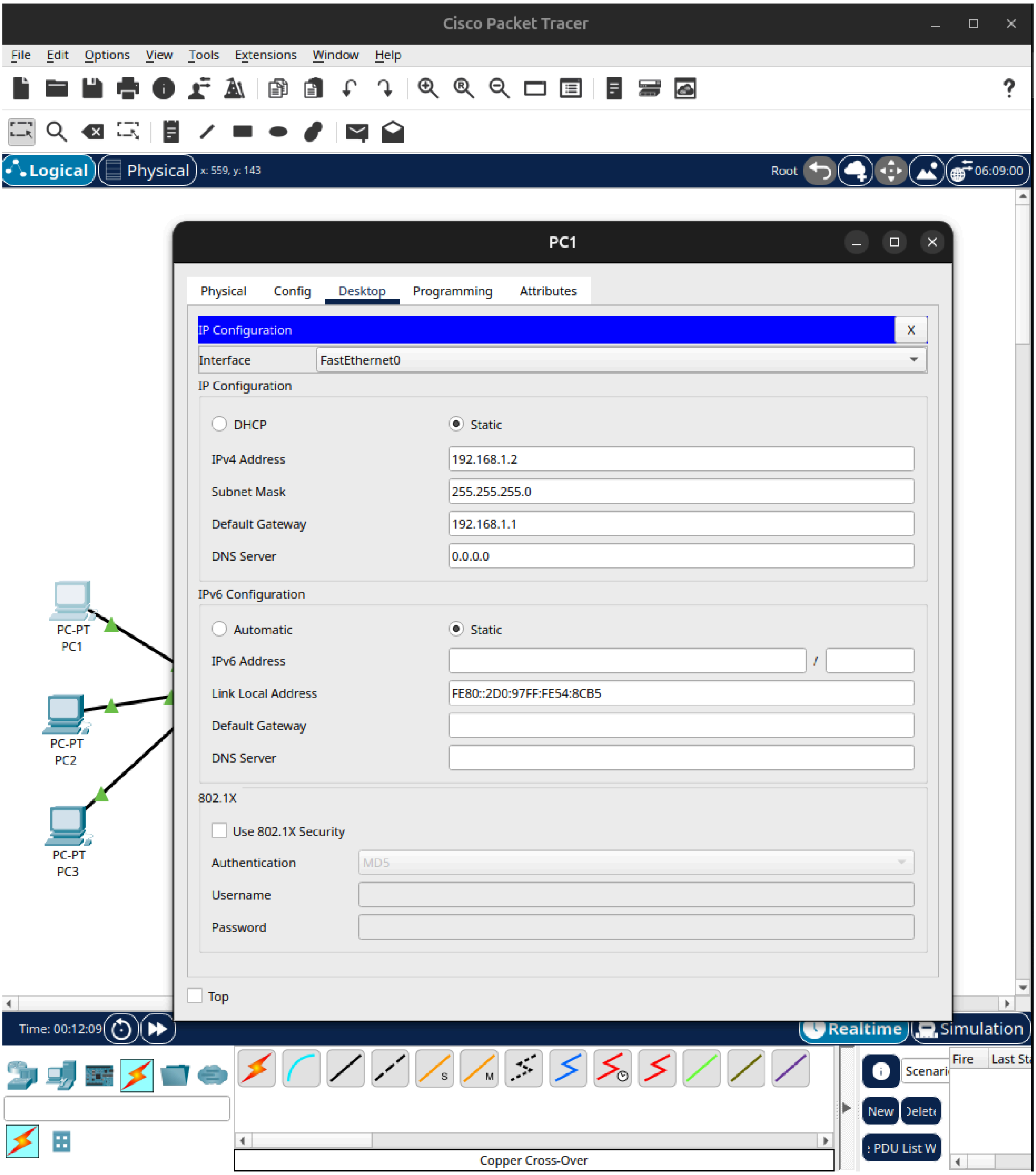
Device	IP Address	Subnet Mask	Default Gateway	Interface
PC1	192.168.1.2	255.255.255.0	192.168.1.1	FastEthernet0
PC2	192.168.1.3	255.255.255.0	192.168.1.1	FastEthernet0
PC3	192.168.1.4	255.255.255.0	192.168.1.1	FastEthernet0
PC4	192.168.2.2	255.255.255.0	192.168.2.1	FastEthernet0
PC5	192.168.2.3	255.255.255.0	192.168.2.1	FastEthernet0
PC6	192.168.3.2	255.255.255.0	192.168.3.1	FastEthernet0
R1 (LAN)	192.168.1.1	255.255.255.0	—	GigabitEthernet0/0
R1 – R2	10.0.12.1	255.255.255.0	—	GigabitEthernet0/1
R1 – R3	10.0.31.2	255.255.255.0	—	GigabitEthernet0/2
R2 – R1	10.0.12.2	255.255.255.0	—	GigabitEthernet0/1
R2 – R3	10.0.23.1	255.255.255.0	—	GigabitEthernet0/2
R2 (PC6)	192.168.3.1	255.255.255.0	—	GigabitEthernet0/0
R3 – R2	10.0.23.2	255.255.255.0	—	GigabitEthernet0/1
R3 (LAN)	192.168.2.1	255.255.255.0	—	GigabitEthernet0/0
R3 – R1	10.0.31.1	255.255.255.0	—	GigabitEthernet0/2

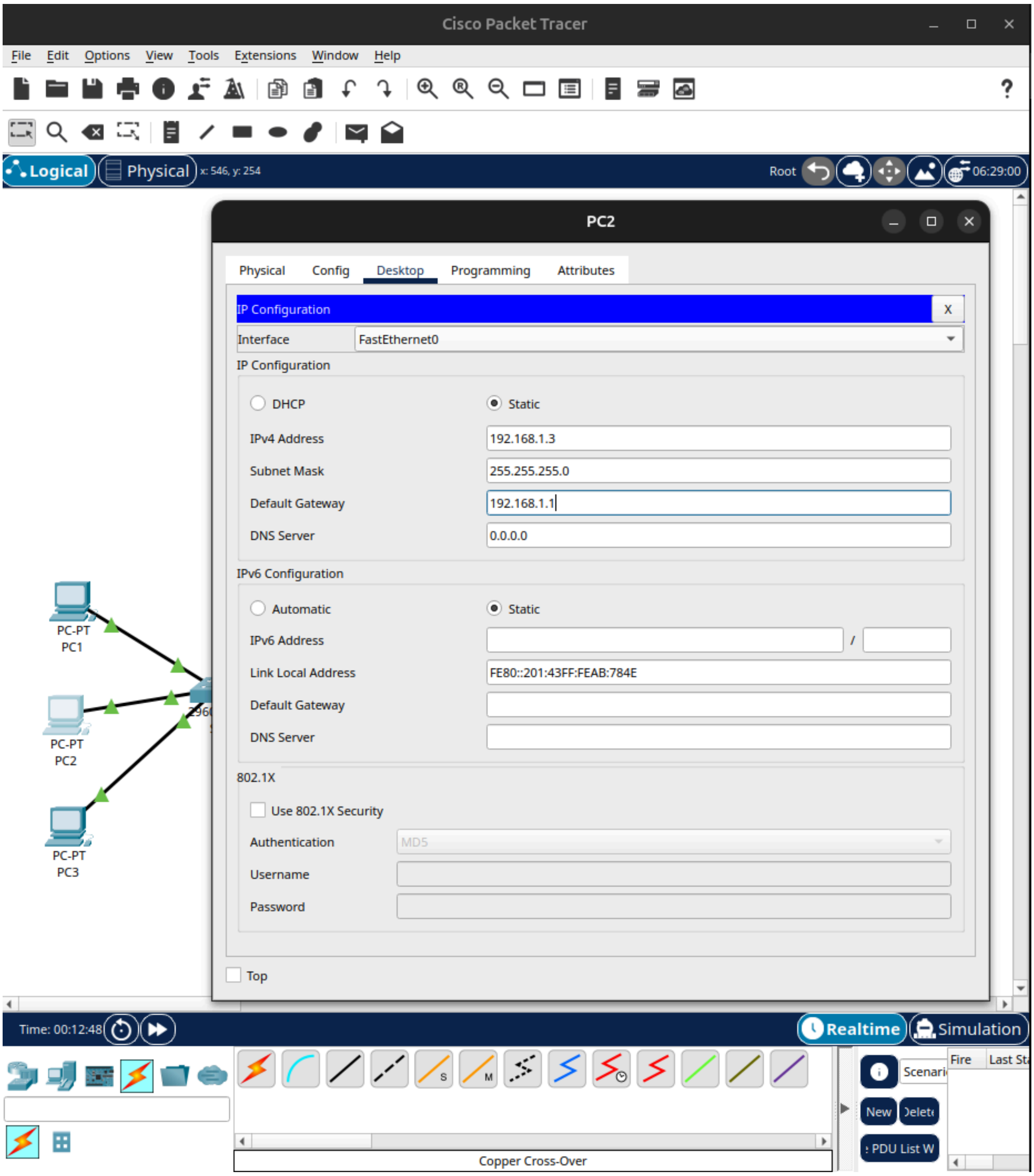
B.2 PC Configuration Steps

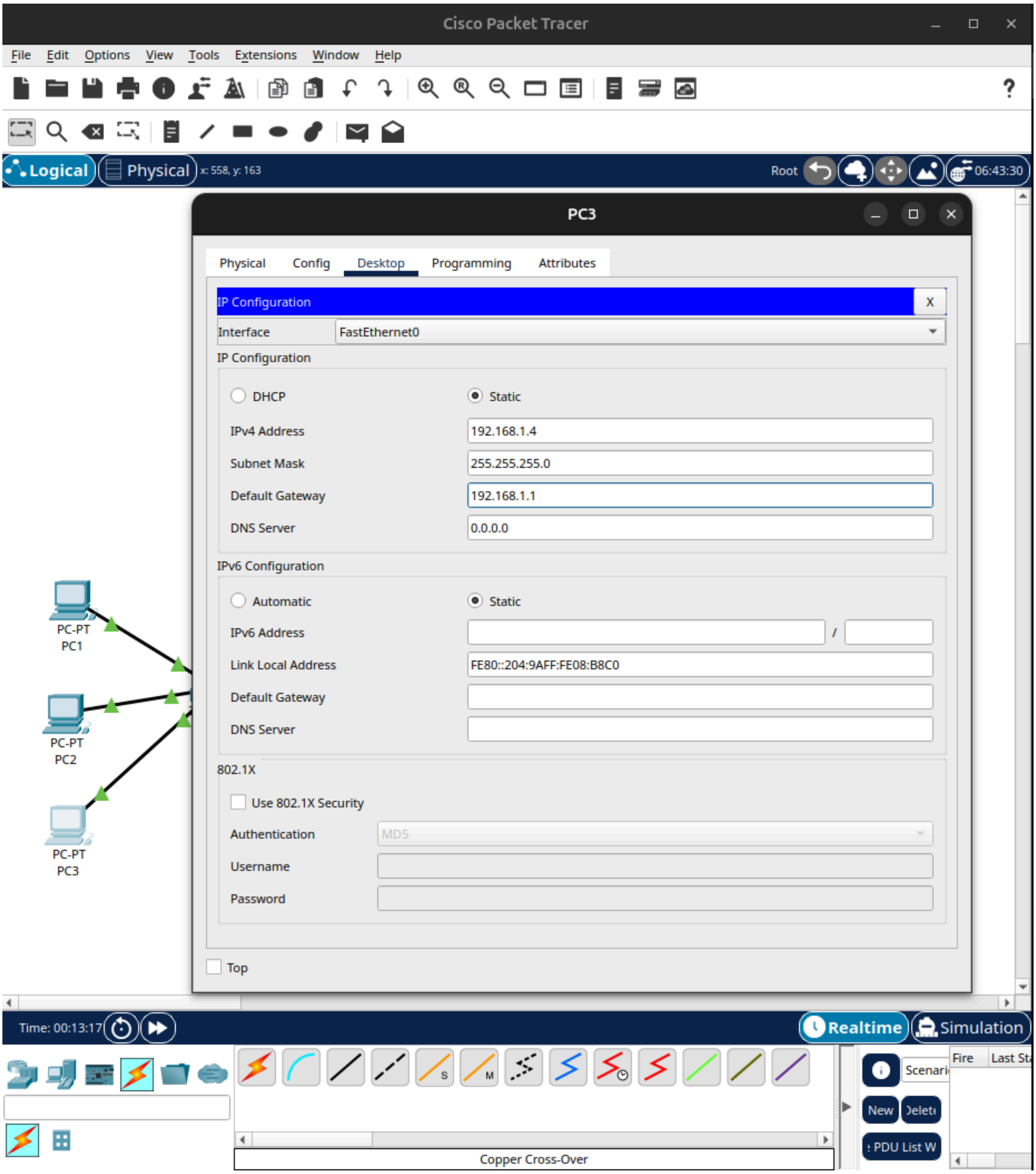
To configure each PC in Cisco Packet Tracer:

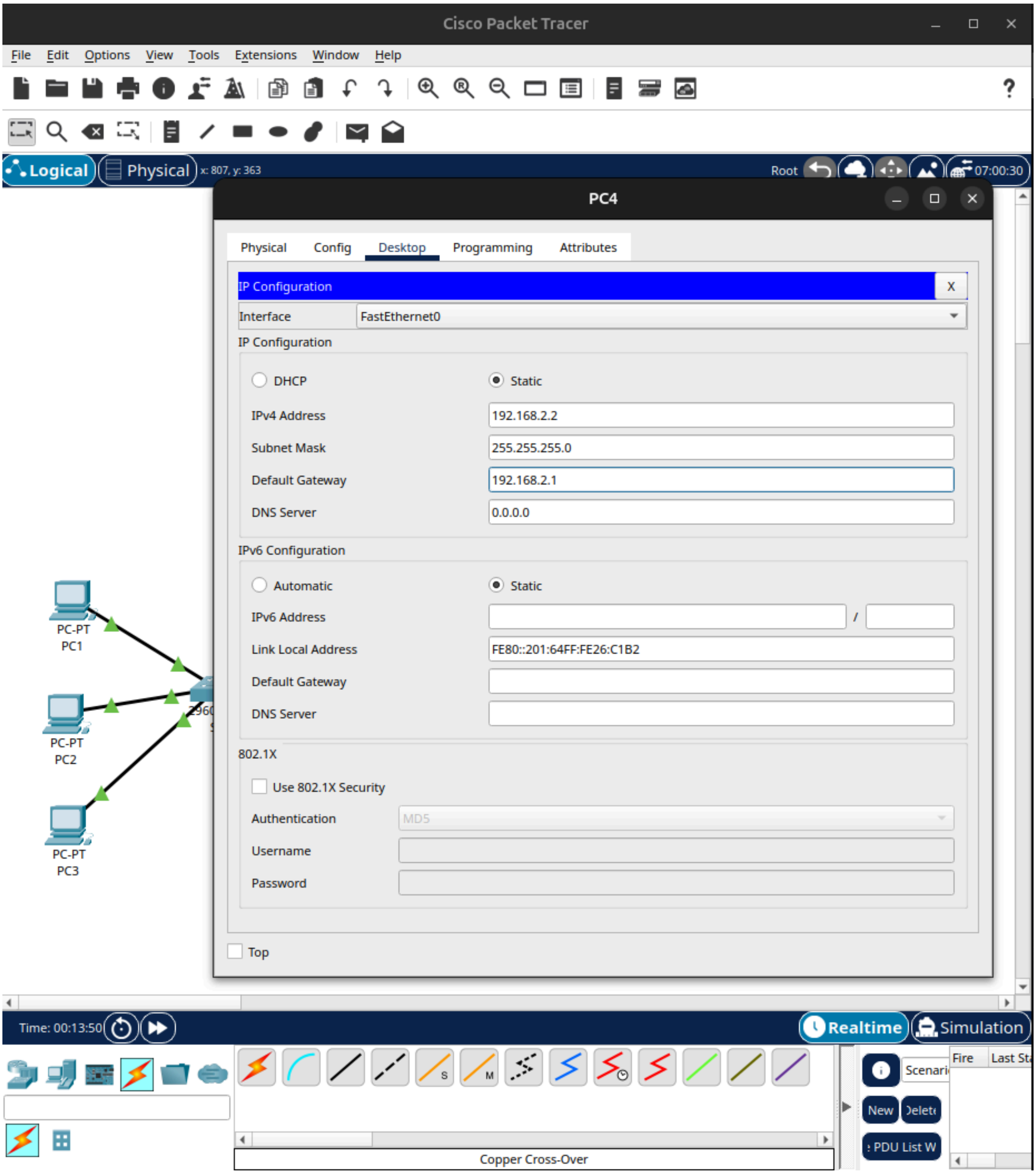
1. Click on the PC → Desktop tab → IP Configuration
2. Select Static radio button
3. Enter IP Address, Subnet Mask, and Default Gateway as per the table above

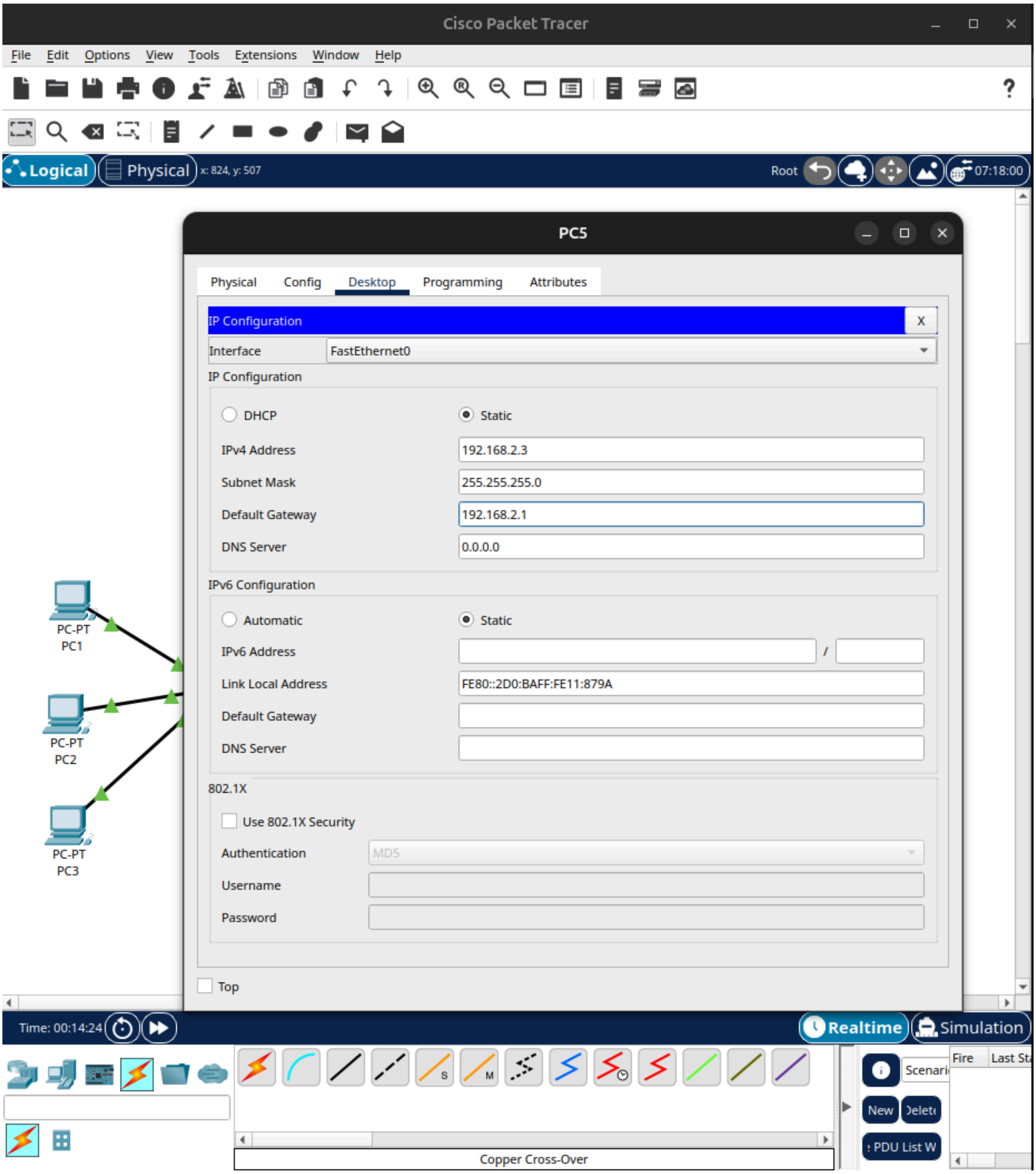
Below are the screenshot of all 6 PCs IP Configuration

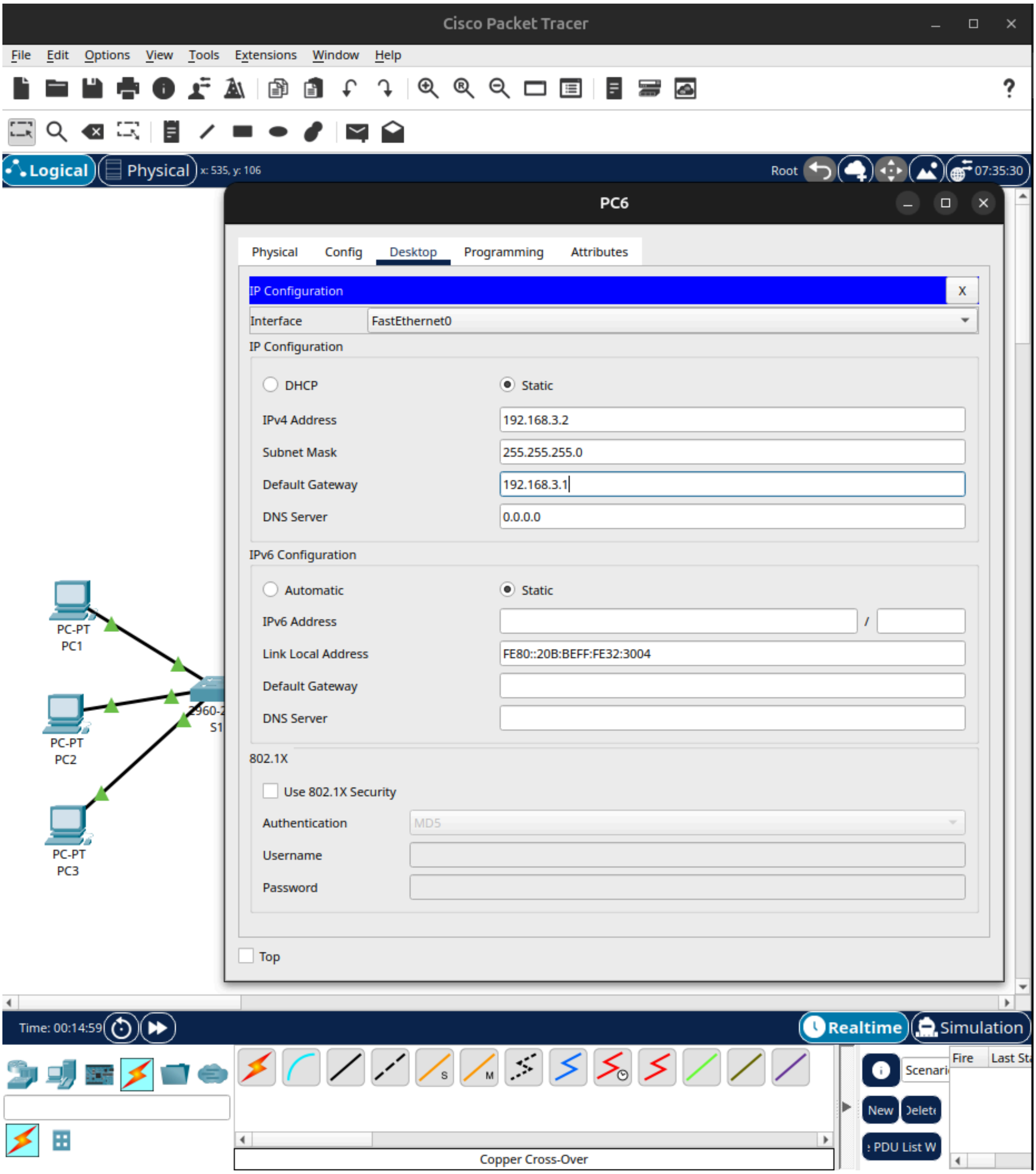



Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 546, y: 254
Root
06:29:00
PC2
Physical Config Desktop Programming Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:43FF:FEAB:784E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:12:48
Realtime
Simulation
Copper Cross-Over
New
PDU List W
Fire
Last St


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 558, y: 163
Root
06:43:30
PC3
Physical Config Desktop Programming Attributes
IP Configuration
X
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.4
Subnet Mask 255.255.255.0
Default Gateway 192.168.1.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::204:9AFF:FE08:B8C0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:13:17
Realtime
Simulation
Copper Cross-Over
Scenari
New
Fire
Last St


Cisco Packet Tracer
PC4
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.2.2
Subnet Mask 255.255.255.0
Default Gateway 192.168.2.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::201:64FF:FE26:C1B2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:13:50
Realtime
Simulation
Copper Cross-Over


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 824, y: 507
Root
07:18:00
PC5
Physical Config Desktop Programming Attributes
IP Configuration
X
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.2.3
Subnet Mask 255.255.255.0
Default Gateway 192.168.2.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2D0:BAFF:FE11:879A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:14:24
Realtime
Simulation
Fire
New
Copper Cross-Over


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 535, y: 106
Root 07:35:30
PC6
Physical Config Desktop Programming Attributes
IP Configuration X
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.3.2
Subnet Mask 255.255.255.0
Default Gateway 192.168.3.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::20B:BEFF:FE32:3004
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
PC-PT PC1
PC-PT PC2
PC-PT PC3
S1
Time: 00:14:59
Realtime Simulation
Copper Cross-Over
New
Fire
Last St

B.3 Router Configuration – R1

Click on R1 → CLI tab and enter the following commands:

Step 1: Enter privileged/config mode

```
Router> enable
Router# config terminal
```

Step 2: Configure interface GigabitEthernet0/0 (LAN side – connected to S1)

```
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Step 3: Configure GigabitEthernet0/1 (WAN – R1 to R2)

```
R1(config)# interface gigabitethernet 0/1
R1(config-if)# ip address 10.0.12.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

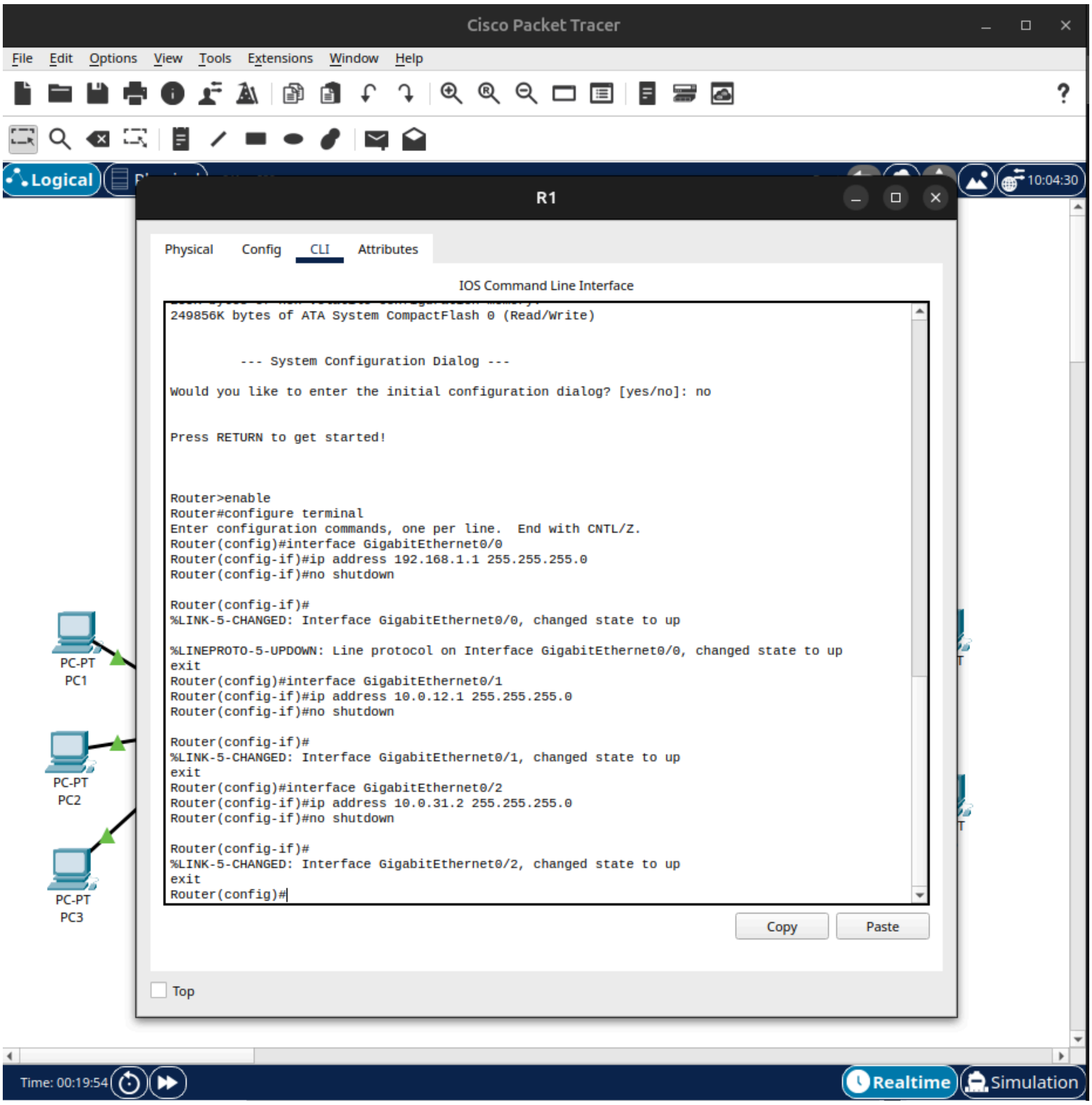
Step 4: Configure GigabitEthernet0/2 (WAN – R1 to R3)

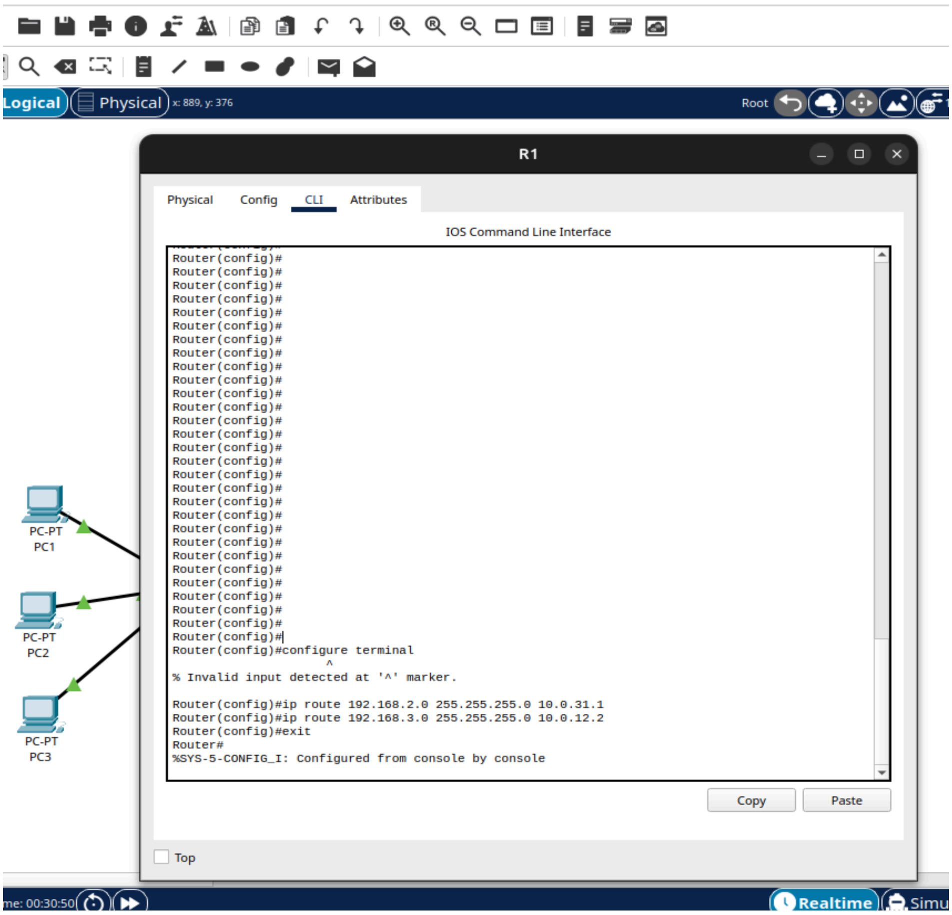
```
R1(config)# interface gigabitethernet 0/2
R1(config-if)# ip address 10.0.31.2 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Step 5: Configure static routes on R1

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.31.1
R1(config)# ip route 192.168.3.0 255.255.255.0 10.0.12.2
R1(config)# exit
```

Screenshot: of R1 CLI


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
10:04:30
R1
Physical Config CLI Attributes
IOS Command Line Interface
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 10.0.12.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 10.0.31.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
Router(config)#
Copy
Paste
Top
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:19:54
Realtime
Simulation



B.4 Router Configuration – R2

Click on R2 → CLI tab and enter:

```
Router> enable
Router# config terminal
```

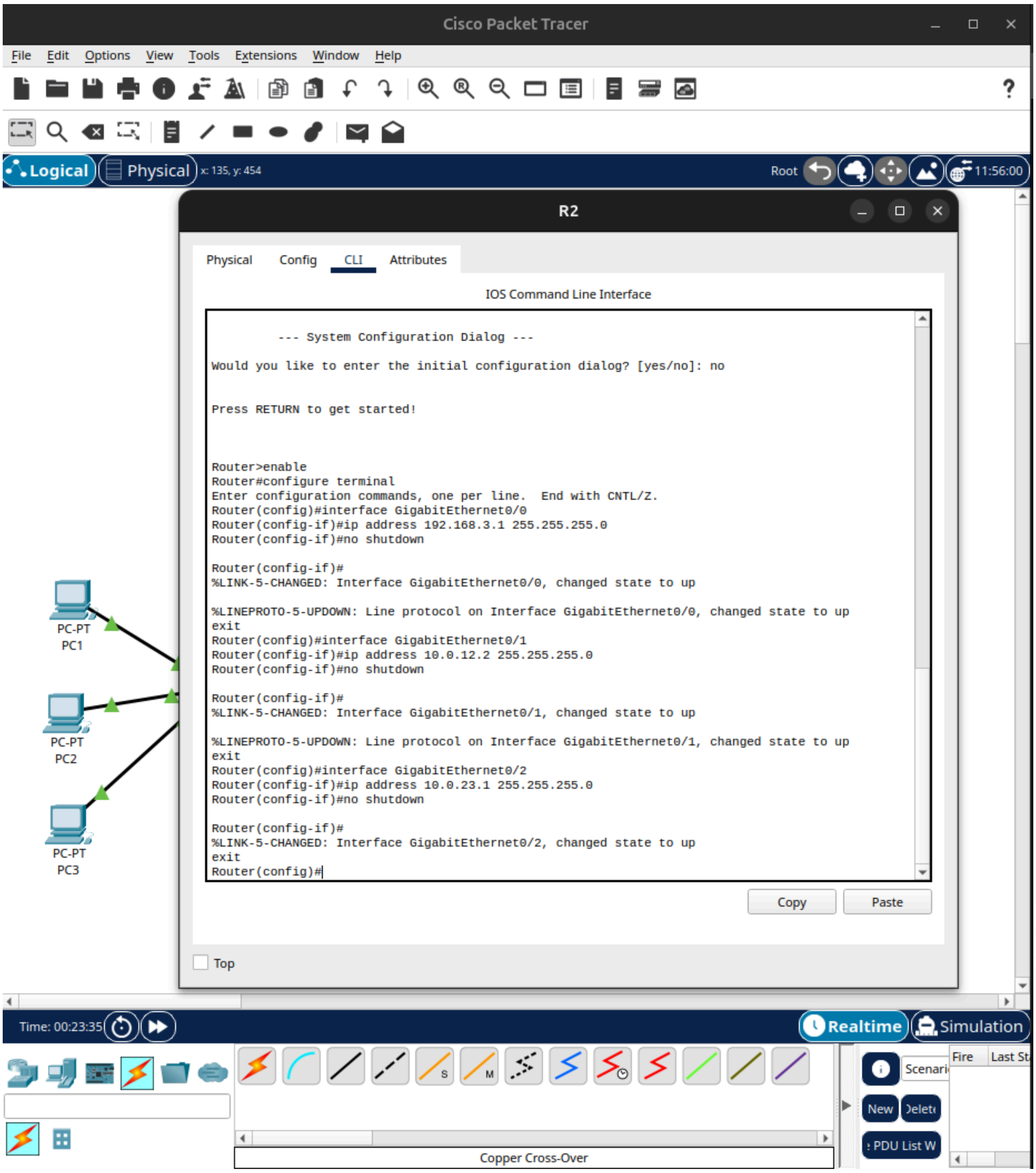
Configure interfaces:

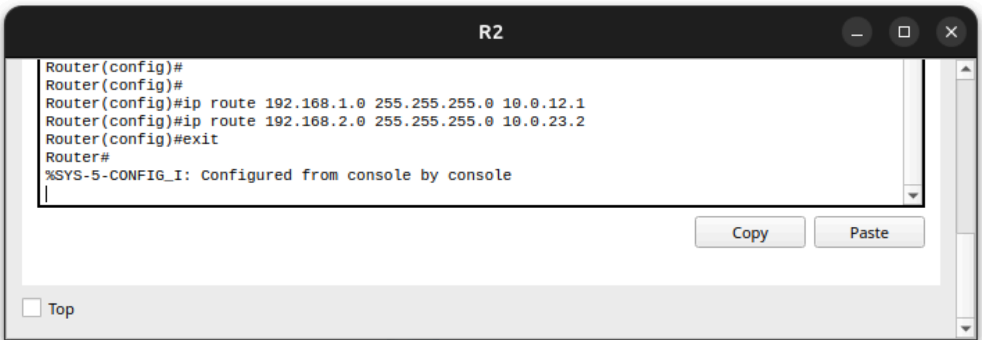
```
R2(config)# interface GigabitEthernet0/0
R2(config-if)# ip address 192.168.3.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface GigabitEthernet0/1
R2(config-if)# ip address 10.0.12.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface GigabitEthernet0/2
R2(config-if)# ip address 10.0.23.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Configure static routes on R2:

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.12.1
R2(config)# ip route 192.168.2.0 255.255.255.0 10.0.23.2
R2(config)# exit
```

Screenshot: R2 CLI





B.5 Router Configuration – R3

Click on R3 → CLI tab and enter:

```
Router> enable
Router# config terminal
Router(config)# hostname R3
```

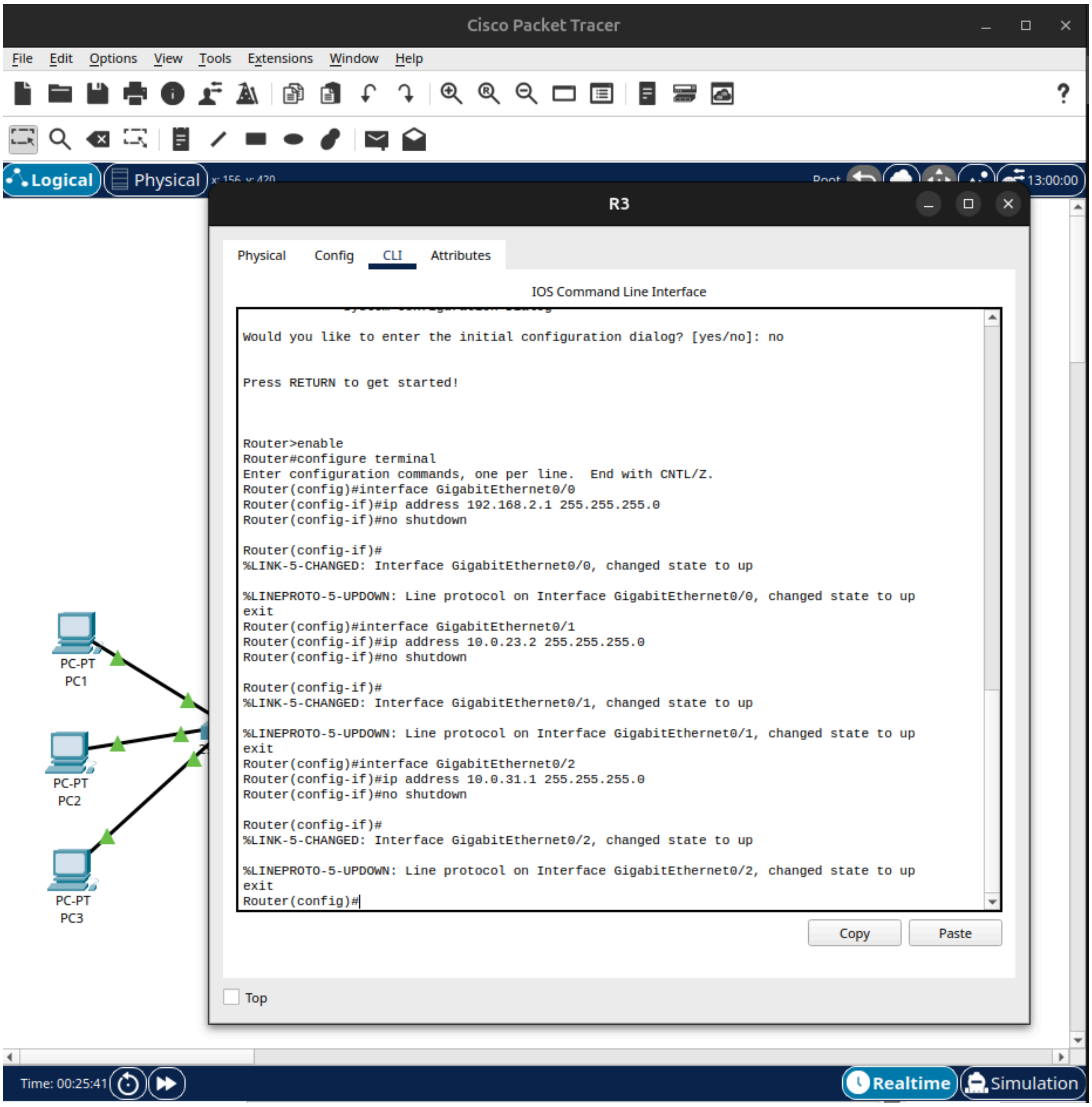
Configure interfaces:

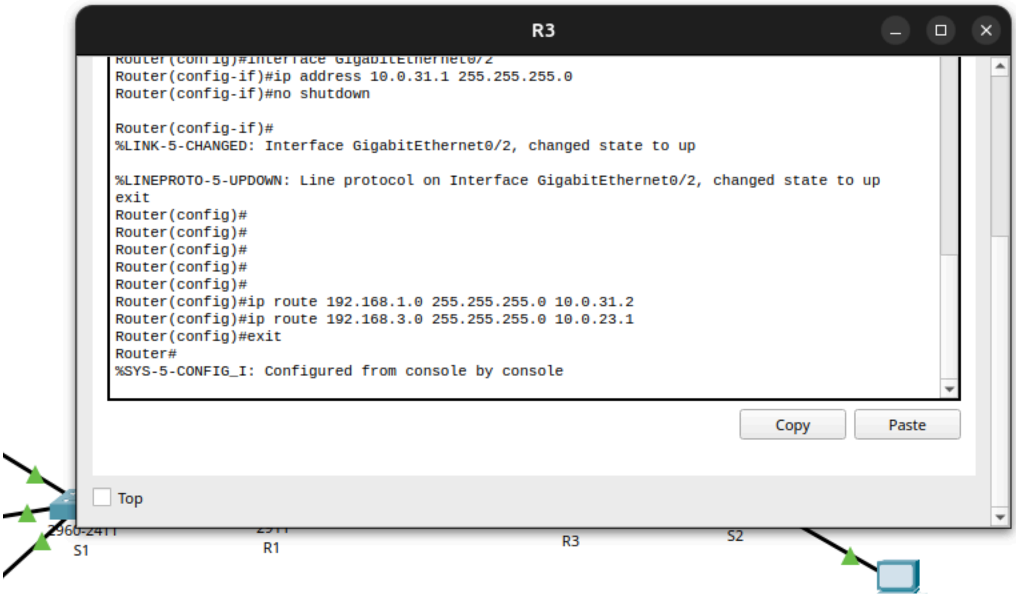
```
R3(config)# interface gigabitethernet 0/0
R3(config-if)# ip address 192.168.2.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)# interface gigabitethernet 0/1
R3(config-if)# ip address 10.0.23.2 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)# interface gigabitethernet 0/2
R3(config-if)# ip address 10.0.31.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
```

Configure static routes on R3:

```
R3(config)# ip route 192.168.1.0 255.255.255.0 10.0.31.2
R3(config)# ip route 192.168.3.0 255.255.255.0 10.0.23.1
R3(config)# exit
```

Screenshot: R3 CLI





Part C: Verification

C.1 Ping Tests

To test connectivity, open the Command Prompt of each source PC (Desktop → Command Prompt) and run the ping command.

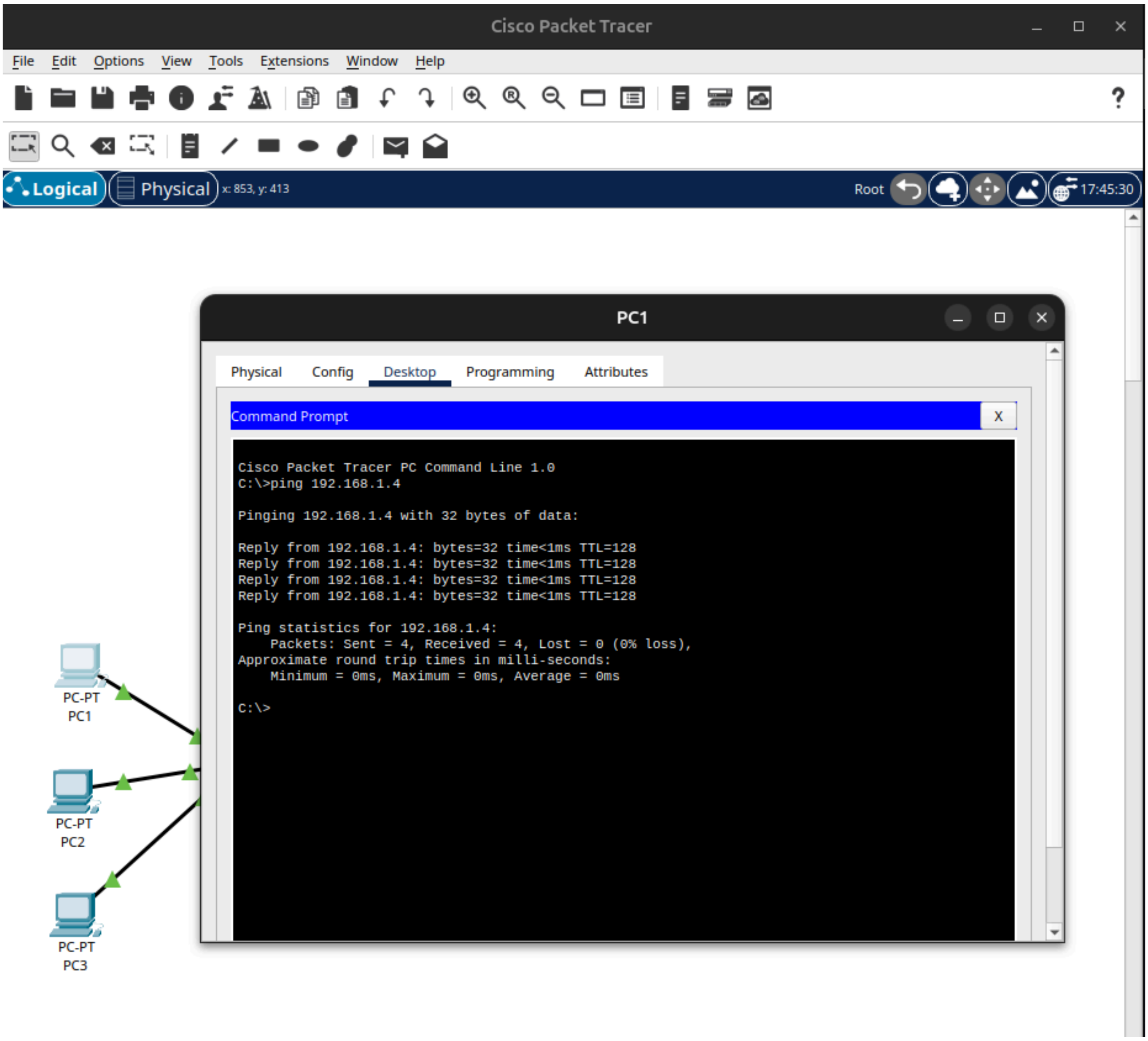
Test 1: PC1 → PC3 (same LAN)

Both PCs are on the same subnet (192.168.1.0/24) connected via S1. The ping should succeed without traversing any router .

```
C:\> ping 192.168.1.4
```

Result: 4 replies received, 0% packet loss

Screenshot:



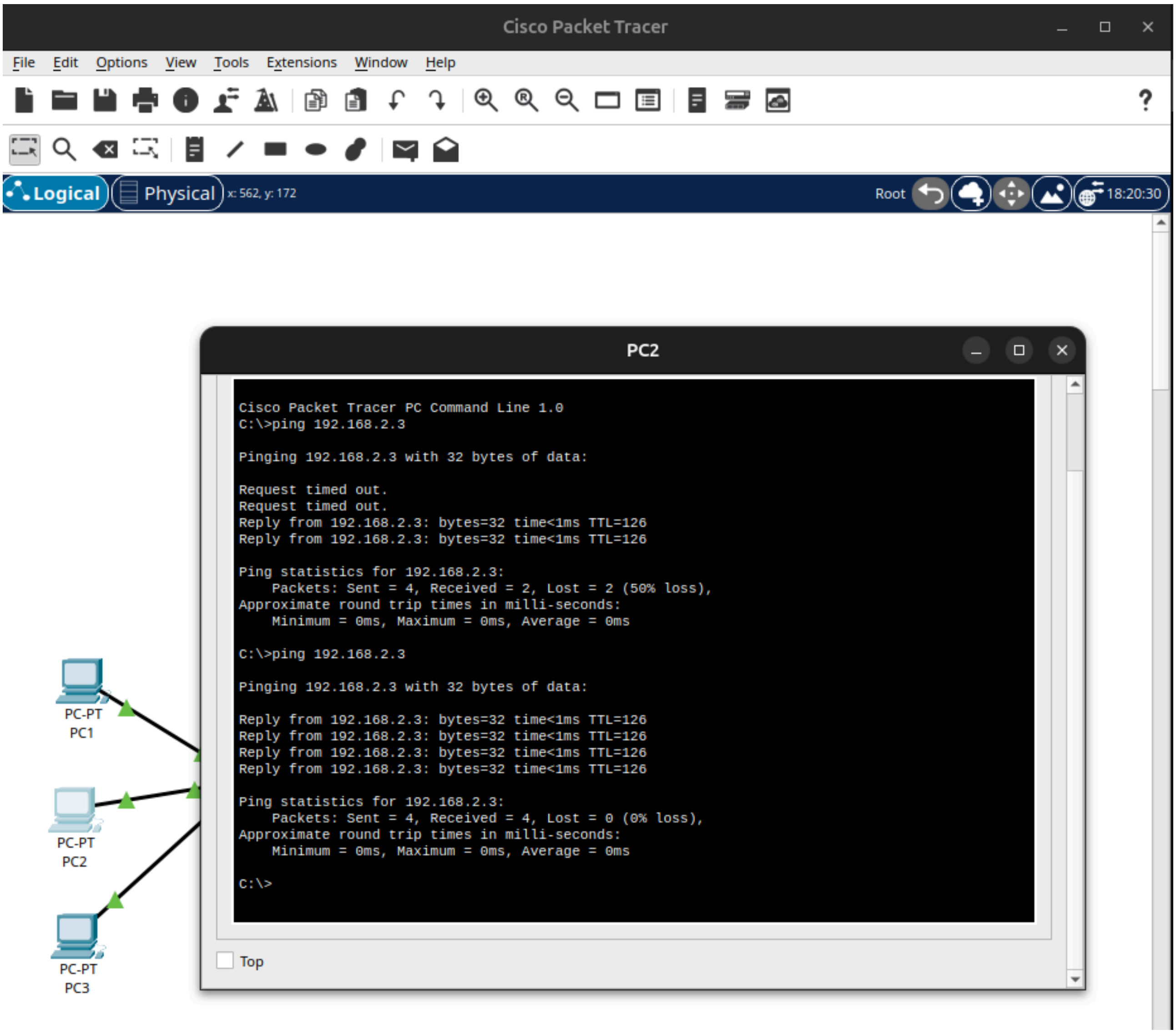
Test 2: PC2 → PC5 (across routers)

PC2 is on LAN 1 (192.168.1.3) and PC5 is on LAN 2 (192.168.2.3). The packet travels: PC2 → S1 → R1 → R3 → S2 → PC5.

```
C:\> ping 192.168.2.3
```

Result: Initially, 50% packet loss (2 replies received) due to the routers performing initial ARP resolution. On the second attempt, 4 replies were received with 0% packet loss (TTL reduced by 2 hops).

Screenshot:



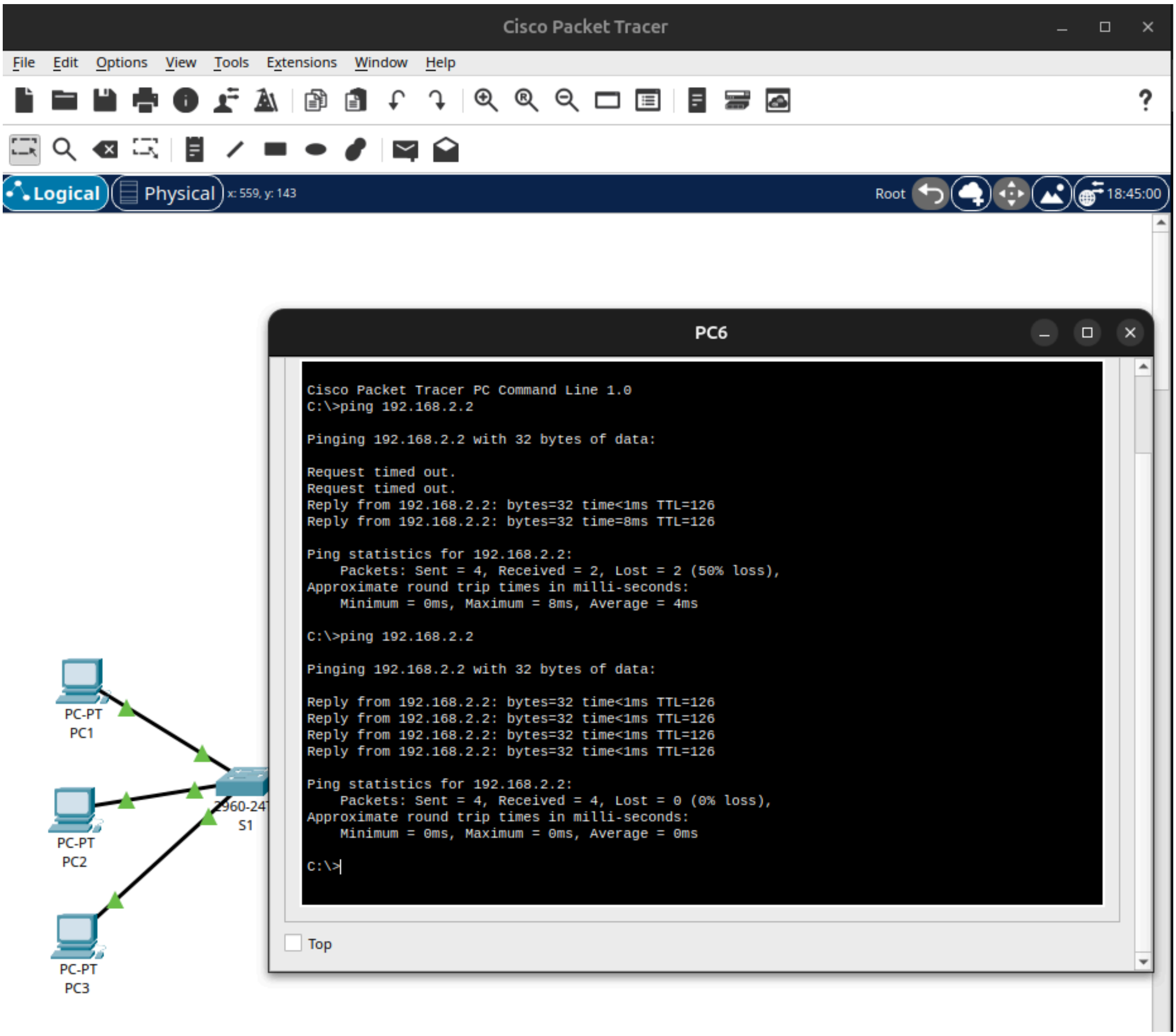
Test 3: PC6 → PC4 (across routers)

PC6 is directly connected to R2 (192.168.3.2) and PC4 is on LAN 2 (192.168.2.2). Packet path: PC6 → R2 → R3 → S2 → PC4.

```
C:\> ping 192.168.2.2
```

Result: Initially, 50% packet loss (2 replies received) due to initial ARP resolution delays. Subsequent ping attempts resulted in 4 replies with 0% packet loss (TTL reduced by 2 hops).

Screenshot:



C.2 Simulation Mode – Packet Flow Analysis

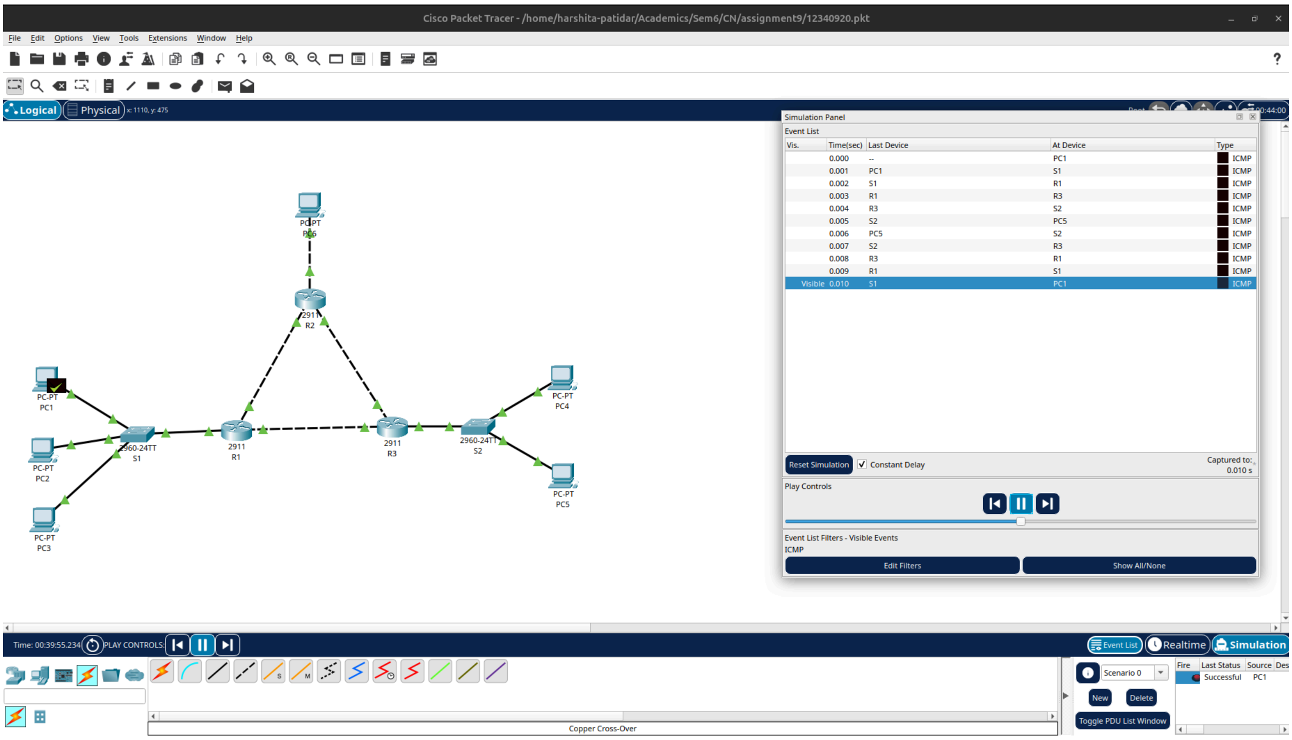
To visually verify the packet flow and routing paths, Packet Tracer's Simulation Mode was utilized. Steps to use Simulation Mode in Cisco Packet Tracer:

1) Switch to Simulation mode using the toggle button at the bottom-right of the screen (Realtime → Simulation).
2) The Event List Filter was edited to show only **ICMP** traffic to isolate the ping request from background network protocols.
3) A Simple PDU (ICMP Echo Request) was generated with **PC1** as the source and **PC5** as the destination.
4) Click 'Play' to observe the packet traveling hop by hop across the network.

Observed packet path for PC1 → PC5:

As shown in the Event List, the packet successfully followed the configured static routes. The forward path traversed: **PC1 → S1 → R1 → R3 → S2 → PC5**. The ICMP Echo Reply successfully navigated the exact reverse path back to PC1, resulting in a successful ping delivery. This confirms that the WAN routing and LAN switching are fully functional and properly integrated.

Screenshot:



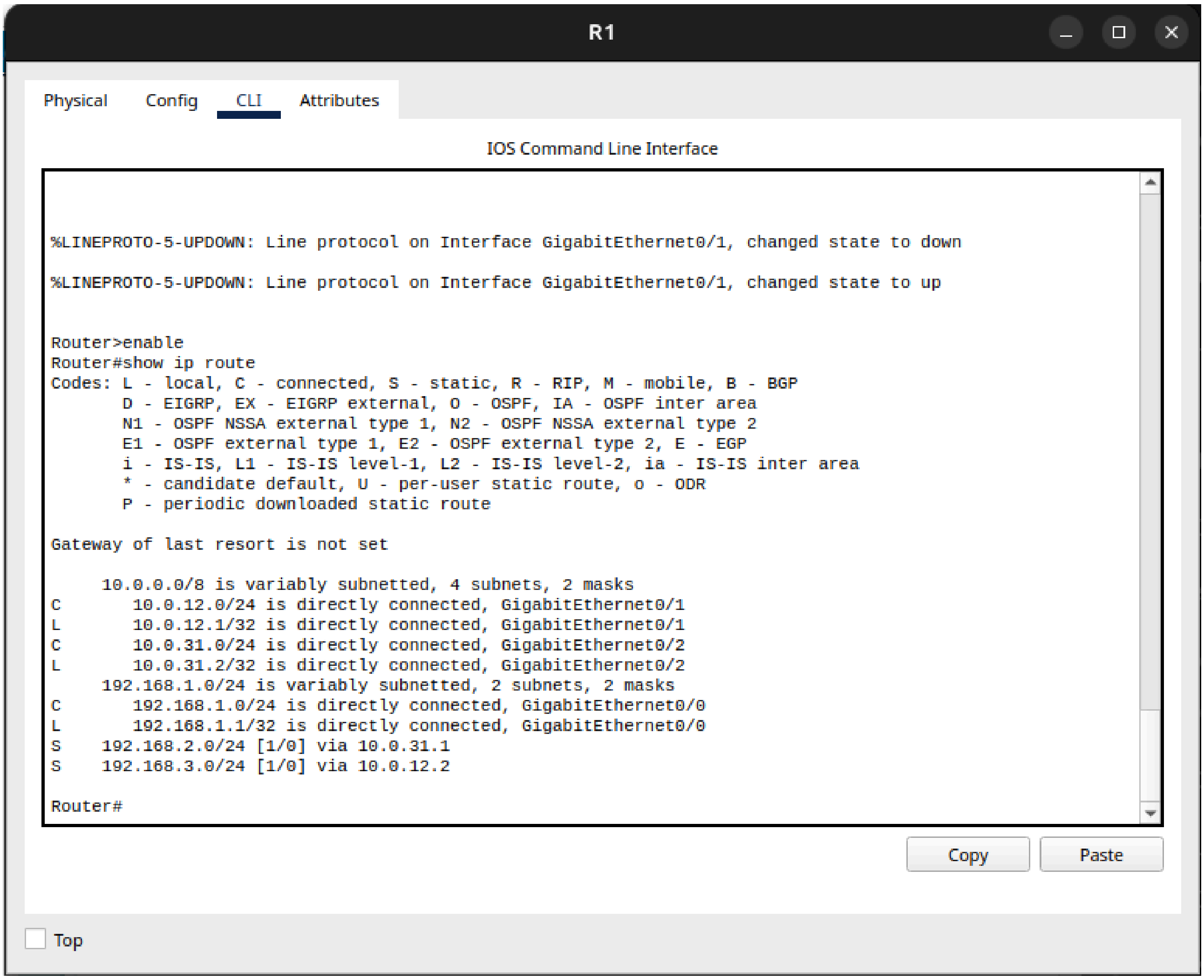
C.3 Verifying Router Configurations

To verify that the static routes were successfully injected into the routing tables, the `show ip route` command was executed in the CLI of each router.

The `show ip route` command output for R1 confirms:

- C / L (Directly Connected / Local Interfaces):
  - `10.0.12.0/24` is directly connected, GigabitEthernet0/1 (WAN to R2)
  - `10.0.31.0/24` is directly connected, GigabitEthernet0/2 (WAN to R3)
  - `192.168.1.0/24` is directly connected, GigabitEthernet0/0 (LAN 1)
- S (Static Routes):
  - `192.168.2.0/24` [1/0] via 10.0.31.1 (Route to LAN 2)
  - `192.168.3.0/24` [1/0] via 10.0.12.2 (Route to PC6)

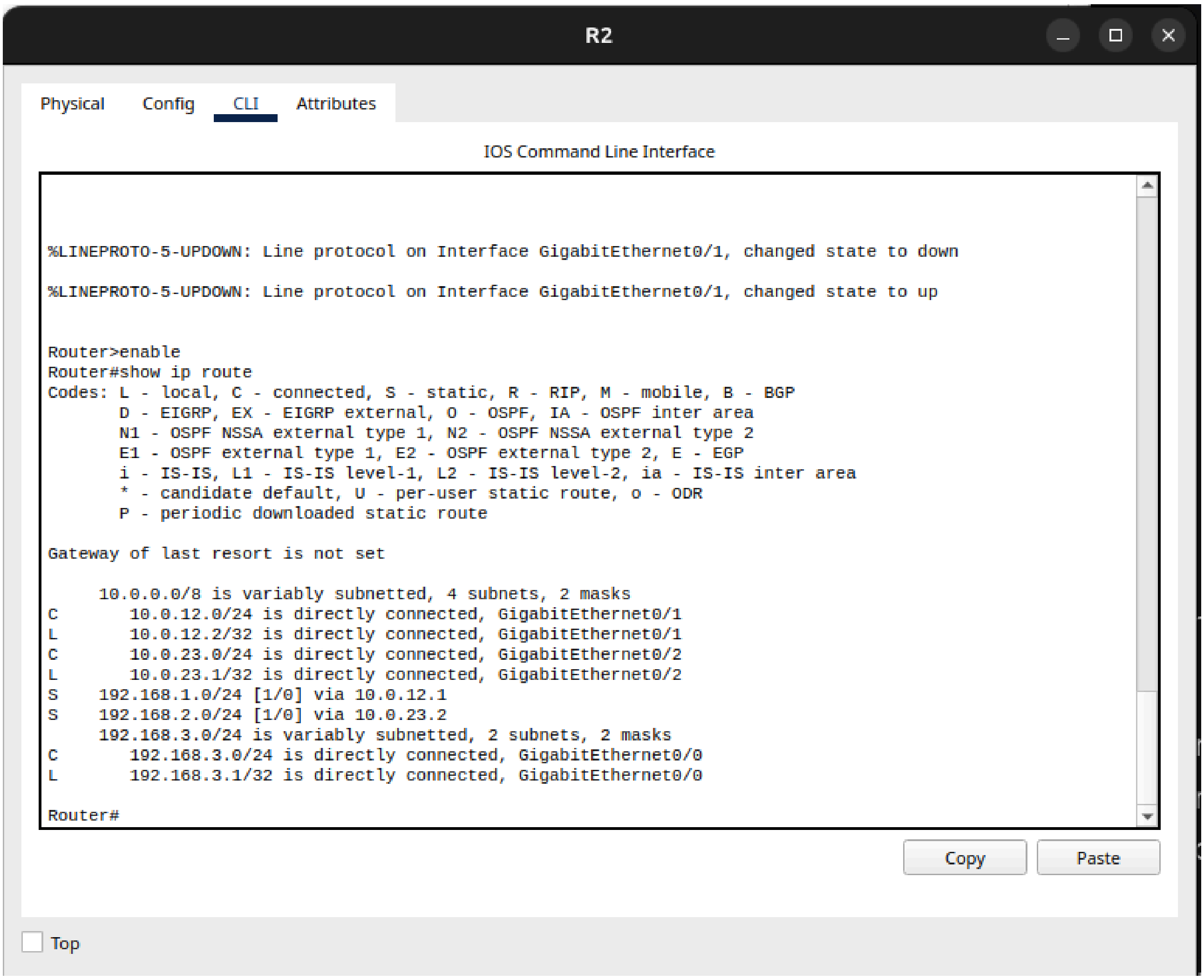
Screenshot:



The `show ip route` command output for R2 confirms:

- C / L (Directly Connected / Local Interfaces):
    - `10.0.12.0/24` is directly connected, GigabitEthernet0/1 (WAN to R1)
    - `10.0.23.0/24` is directly connected, GigabitEthernet0/2 (WAN to R3)
    - `192.168.3.0/24` is directly connected, GigabitEthernet0/0 (PC6)
- S (Static Routes):
    - `192.168.1.0/24` [1/0] via 10.0.12.1 (Route to LAN 1)
    - `192.168.2.0/24` [1/0] via 10.0.23.2 (Route to LAN 2)

Screenshot:



The `show ip route` command output for R3 confirms:

- C / L (Directly Connected / Local Interfaces):
  - `10.0.23.0/24` is directly connected, GigabitEthernet0/1 (WAN to R2)
  - `10.0.31.0/24` is directly connected, GigabitEthernet0/2 (WAN to R1)
  - `192.168.2.0/24` is directly connected, GigabitEthernet0/0 (LAN 2)
- S (Static Routes):
  - `192.168.1.0/24` [1/0] via 10.0.31.2 (Route to LAN 1)
  - `192.168.3.0/24` [1/0] via 10.0.23.1 (Route to PC6)

Screenshot: